

LAB EXPERIMENT RECORD

Set 10: Survey Responses Analysis

Department	AIDS
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
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Date	06/10/2026
Day	Wednesday

AIM

To analyze survey responses using **R and Tableau** by creating a grouped bar chart, stacked bar chart, response table, and an interactive dashboard.

Student_ID	Gender	Age	Course	Study_Time (hrs)	Videos_Watched	Quiz_Score	Login_Date
L01	Male	20	R	3.5	12	78	2025-01-05
L02	Female	22	R	4.2	15	85	2025-01-05
L03	Male	19	SQL	2.0	8	65	2025-02-08
L04	Female	21	R	5.0	18	92	2025-02-08
L05	Male	23	R	2.5	9	70	2025-03-12
L06	Female	20	SQL	4.0	14	88	2025-03-12

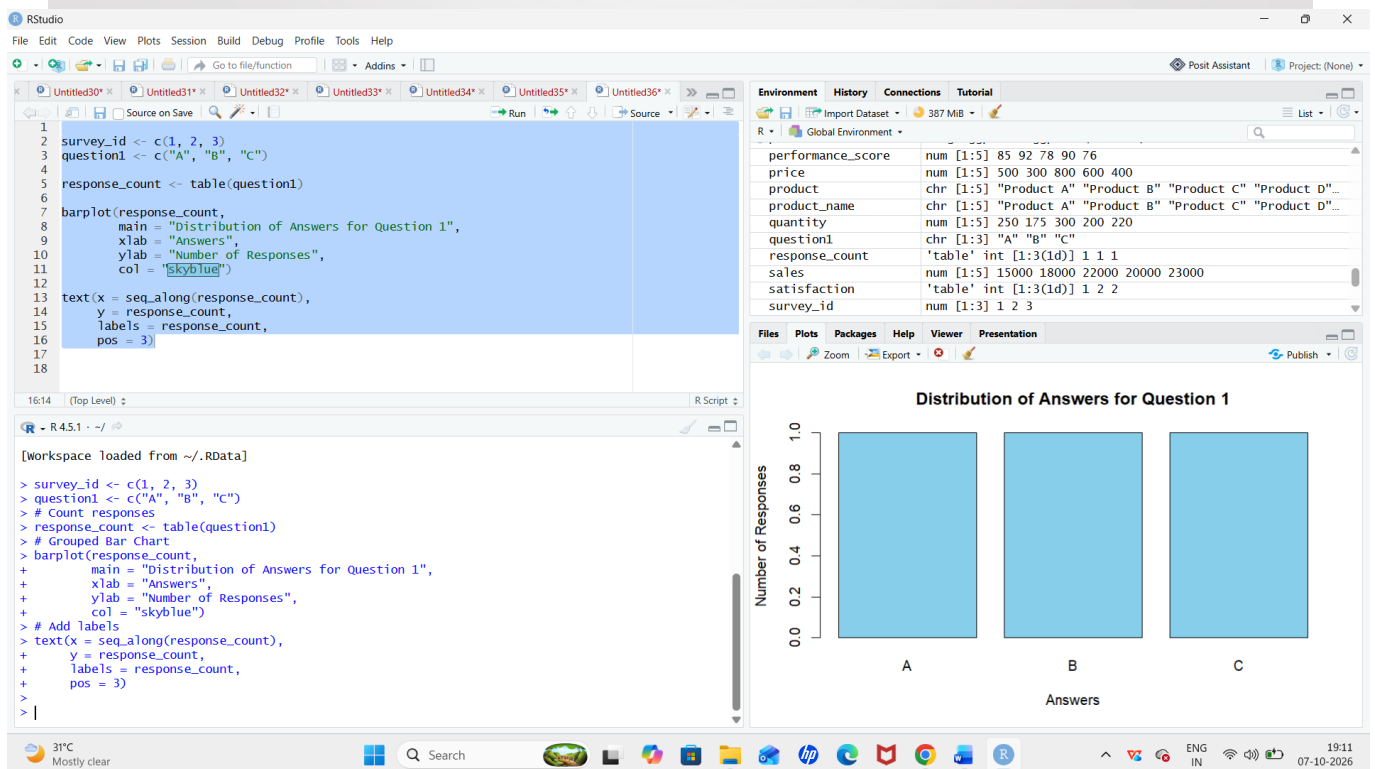
1. Create a grouped bar chart to visualize the distribution of answers for Question

```
survey_id <- c(1, 2, 3)
question1 <- c("A", "B", "C")

response_count <- table(question1)

barplot(response_count,
        main = "Distribution of Answers for Question 1",
        xlab = "Answers",
        ylab = "Number of Responses",
        col = "skyblue")

text(x = seq_along(response_count),
     y = response_count,
     labels = response_count,
     pos = 3)
```

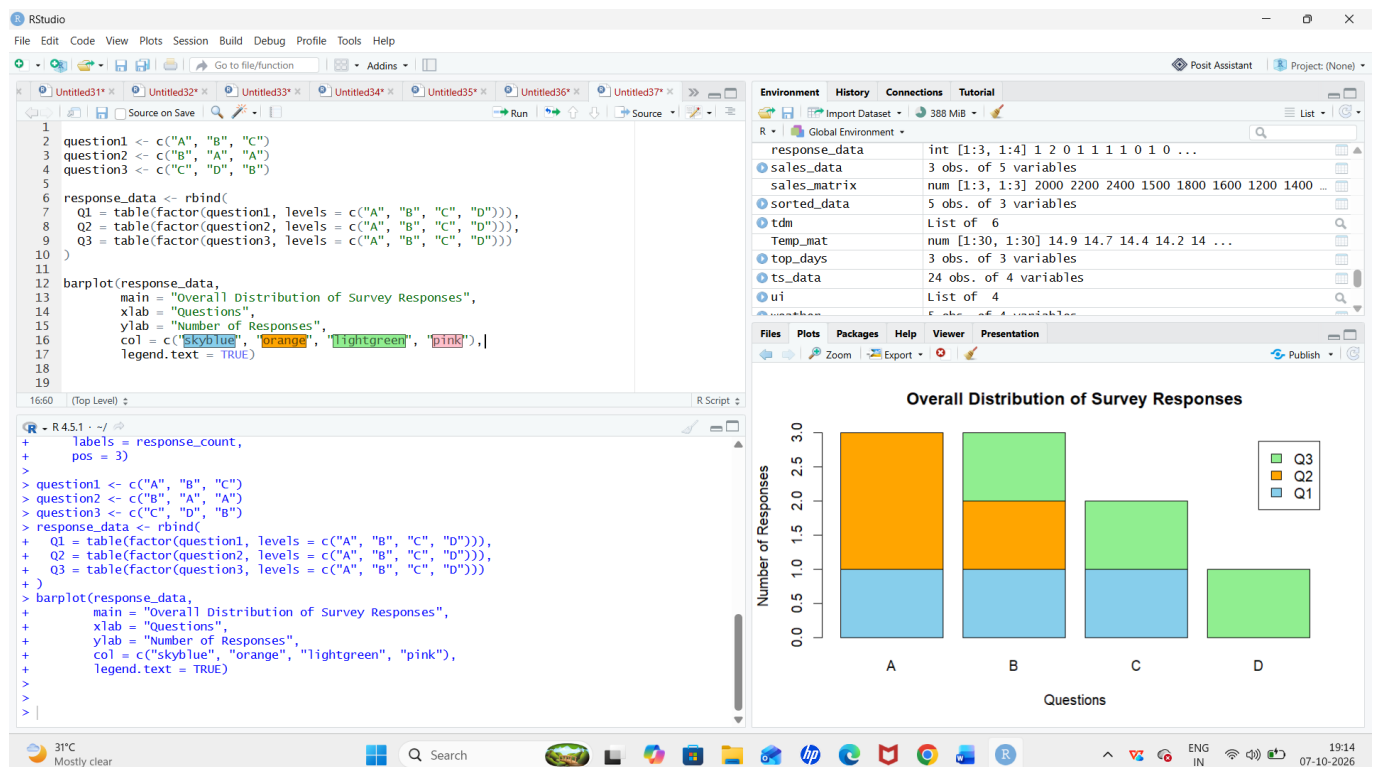


2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

```
question1 <- c("A", "B", "C")
question2 <- c("B", "A", "A")
question3 <- c("C", "D", "B")

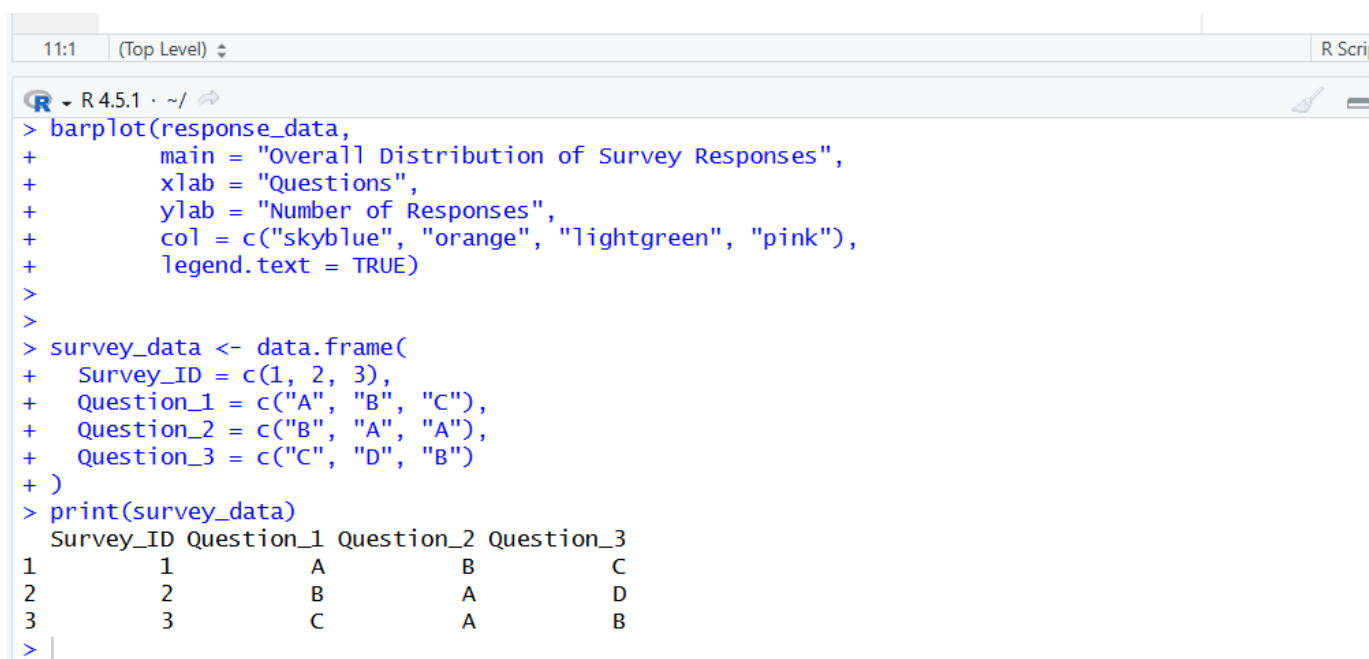
response_data <- rbind(
  Q1 = table(factor(question1, levels = c("A", "B", "C", "D"))),
  Q2 = table(factor(question2, levels = c("A", "B", "C", "D"))),
  Q3 = table(factor(question3, levels = c("A", "B", "C", "D")))
)

barplot(response_data,
  main = "Overall Distribution of Survey Responses",
  xlab = "Questions",
  ylab = "Number of Responses",
  col = c("skyblue", "orange", "lightgreen", "pink"),
  legend.text = TRUE)
```



3 Using R programming, convert Login_Date into Date format..

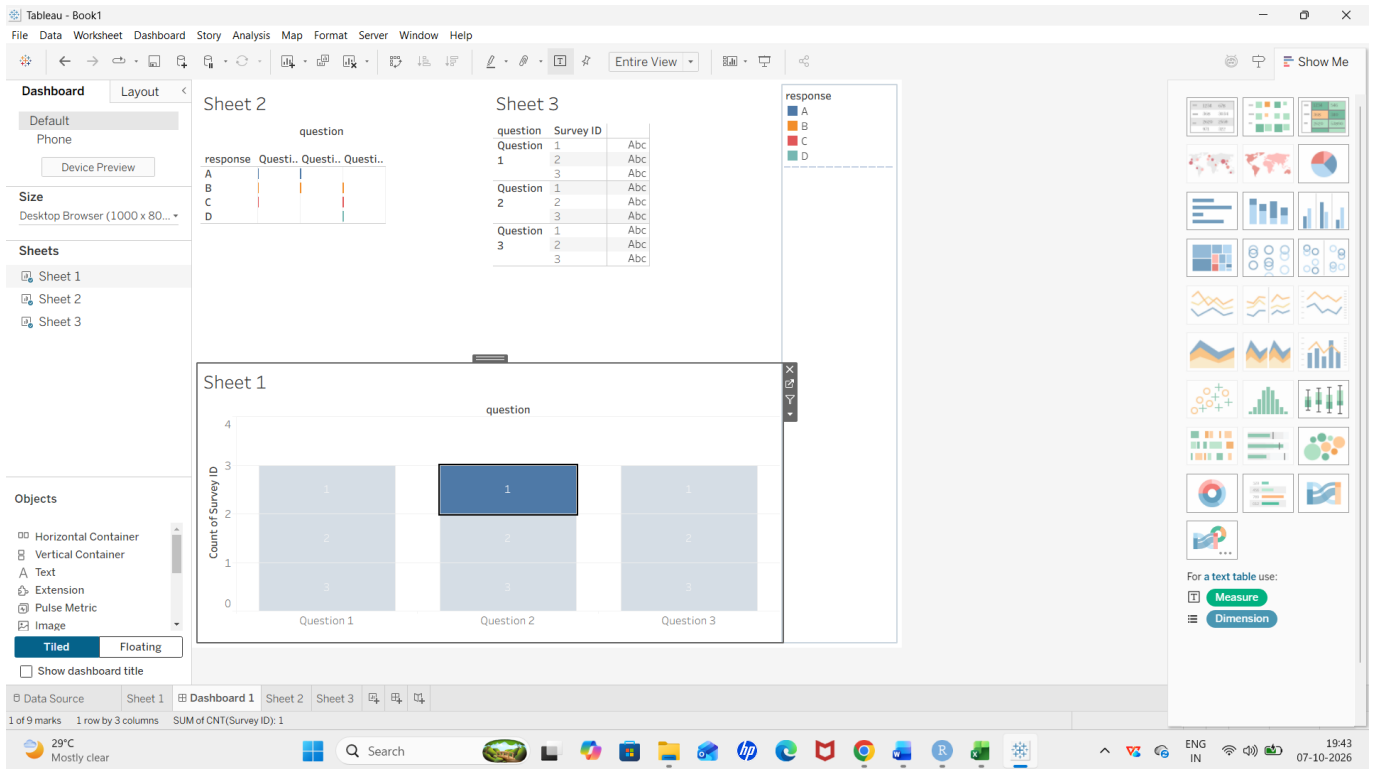
```
survey_data <- data.frame(  
  Survey_ID = c(1, 2, 3),  
  Question_1 = c("A", "B", "C"),  
  Question_2 = c("B", "A", "A"),  
  Question_3 = c("C", "D", "B")  
)  
  
print(survey_data)
```



The screenshot shows an R Studio console window. The title bar indicates the version is R 4.5.1. The console output shows the execution of several R commands. First, a barplot is created for 'response_data' with a main title 'Overall Distribution of Survey Responses', x-axis label 'Questions', y-axis label 'Number of Responses', and four colored bars (skyblue, orange, lightgreen, pink). Then, a data frame 'survey_data' is created with three columns: 'Survey_ID', 'Question_1', 'Question_2', and 'Question_3'. The data frame is printed, showing the following table:

	Survey_ID	Question_1	Question_2	Question_3
1	1	A	B	C
2	2	B	A	D
3	3	C	A	B

14.



PROCEDURE

1. Enter the survey response data containing Survey ID, Question 1, Question 2, and Question 3.
2. Create a grouped bar chart to show the distribution of answers for Question 1.
3. Create a stacked bar chart to represent the overall distribution of responses across all three questions.
4. Create a table to display the responses given by each survey participant.
5. Import the dataset into Tableau.
6. Create the required charts and table in separate worksheets.
7. Combine the worksheets into a Tableau dashboard.
8. Add labels and interactive filters for better data exploration.

RESULT

The survey responses were successfully analyzed using R and Tableau. The grouped bar chart shows the distribution of Question 1 answers, the stacked bar chart shows responses across all questions, and the table displays individual survey responses. An interactive Tableau dashboard was also created to provide a clear view of the survey results

